ALEKYA LINGA

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TECHNICAL SKILLS

- **Programming Languages**: C, Java, Python.
- **Databases**: MySQL, Oracle SQL.
- **Web Technologies**: HTML5, CSS3, JavaScript.
- **Software and Tools**: Git, VSCode, Eclipse, Linux.
- **Cloud Services**: AWS, Azure DevOps
- **Data Engineering Tools:** SSIS, Tableau

EDUCATION

Master of Computer Science Aug 2023 – Dec 2024
Kennesaw State University, Marietta, Georgia.

Bachelor of Technology, Computer Science Aug 2019 – Apr 2023
Vasireddy Venkatadri Institute of Technology, Guntur, India.

ACADEMIC PROJECTS

Blockchain-Based Voting System May 2024 – Jul 2024
Kennesaw State University, Marietta, Georgia
Technology: Multichain, python

- Developed a secure and transparent electronic voting system using blockchain technology to replace traditional paper ballots, ensuring the integrity and accessibility of the electoral process.
- Designed the system architecture including blockchain network setup, smart contract development, and user interfaces to create a tamper-resistant voting platform that records votes immutably on a blockchain.
- Utilized Python, blockchain frameworks, and tools like Gradio for the user interface, along with consensus mechanisms to ensure decentralized vote validation and counting.
- Successfully demonstrated a prototype that allows secure, anonymous, and verifiable voting via computers or smartphones, with the potential to scale for national elections, contributing to the modernization of electoral systems.

Intrusion Detection System Jan 2023 – Mar 2023
Vasireddy Venkatadri Institute of Technology, Guntur, India.
Technology: Python, Google Colab.
Libraries: NumPy, Pandas, matplotlib, seaborn, sklearn, SciPy.

- Developed a Machine Learning-based Intrusion Detection System to enhance security monitoring.
- Designed an ML model that analyses network traffic to detect anomalies and potential security breaches.
- Pre-processed large datasets to remove noise and improve detection accuracy.
- Conducted comparative analysis of ML models to determine highest detection accuracy, ensuring system reliability.
- Integrated the model with Google Colab for experimentation, visualization, and performance improvement.
- Designed user-friendly dashboards for data visualization and monitoring of intrusion patterns.

Managing Traffic System May 2022 – Dec 2022

Vasireddy Venkatadri Institute of Technology, Guntur, India.

Technology: Java, XML, MySql

- It is an Android-based application that guides the user by providing traffic police details at that location and also specifies rush details.
- The main objective of the Managing Traffic System is to manage the details of Vehicle Types, Routes, Length, Traffic Polices, Traffic.
- It is an organization, arrangement, guidance and control of both stationary and moving traffic, including pedestrians, bicyclists and all types of vehicles.
- This project was selected as one of the top 5 projects in our department's projects expo.

INTERNSHIP

Research Assistantship April 2024 – Dec 2024

- Worked as a research assistant at Kennesaw State University, conducted research under the mentorship of Professor Sarah North, gaining valuable academic and technical insights.
- Focused on enhancing firewall security by integrating **dynamic anomaly detection** techniques to identify and mitigate potential threats in real time.
- Utilized **machine learning algorithms** to analyze network traffic patterns and detect anomalies that could indicate cyber threats such as malware, intrusion attempts, or DDoS attacks.

PUBLICATIONS

Accurate Stress Detection of Elderly Patients Using ML Algorithms Sep 2022 – Feb 2023

- The paper presents a novel approach to detecting stress levels in elderly patients using K-Nearest Neighbors (KNN) and Naive Bayes classifiers, achieving high accuracy in classification based on physiological data.
- Introduced a data-driven methodology for real-time stress detection, incorporating advanced machine learning techniques to improve accuracy and reliability.
- Provided a comprehensive analysis of the model's performance using real-world data, highlighting its potential for integration into healthcare monitoring systems.

CERTIFICATIONS

- AWS Academy Cloud Foundations - AWS Academy Graduate
- Cloud Computing – NPTEL
- JAVA Full Stack – Wipro Talent NeXT
- Python Functions, Files, and Dictionaries – Coursera
- Python Classes and Inheritance - Coursera